PLTW POE

Activity 1.1.2 What’s Simple About a Combine?

PLTW

WHAT’S SIMPLE ABOUT A COMBINE?

ACTIVITY SUPPORTS

The Combine Harvester

GOALS	MATERIALS	RESOURCES

- Identify six types of simple machines.
- Calculate mechanical advantage.
- Replicate simple machines using VEX® components.

GOALS	MATERIALS	RESOURCES

- VEX® V5 POE Kit

GOALS	MATERIALS	RESOURCES

- **Simple Machines**
- **VEX® Parts**
- **VEX® V5 Build Support Guides**
- **Activity 1.1.2 What's Simple About a Combine? (Downloadable PDF)**

> In the early 1800s, 90 percent of the United States population lived and worked on farms. Today, over 200 years later, only 2 percent of the population lives and works this way. One of the driving forces behind this change in society is automated farming equipment.

The combine harvester, or simply the combine, is an automated farming machine that combines the tasks of harvesting, threshing, separating, and cleaning grain crops on a farm. This and other automated farming machines make each agricultural worker vastly more productive.

In this activity, you will learn how the combine harvester uses several simple machines to complete its tasks. You also learn how to calculate the mechanical advantage of each simple machine. To expand your understanding, you end the activity using your VEX® V5 POE Kit to create some simple machines used in the combine harvester.

Simple Machines

Inside the combine harvester, several simple machines work together to harvest, thresh, and clean crops on a farm. Before learning how the combine works, you must build a solid foundation of knowledge about simple machines.

1. To learn about simple machines, the creation and calculation of mechanical advantage, and more, review **Simple Machines**.

2. Watch the video to look at the simple machines inside a combine harvester as it harvests corn.

How a Combine Works video[1]

Reflection

Name and describe the purpose of each simple machine in the combine harvester.

Mechanical Advantage

Each simple machine has the potential to create mechanical advantage in certain scenarios. Consider the following problems to understand how and why simple machines are so useful.

Engineering Notebook

Document your calculations and drawings for each problem.

3

A farmer uses a wheelbarrow to haul a 240 lb load of grain. The length from the wheel and axle to the center of the load is 2 ft. The length from the wheel and axle to the effort is 6 ft.

- Illustrate and annotate the lever system described above.
- Determine the ideal mechanical advantage (IMA) of the system.

Reflection

Was the weight of the load of grain useful as you solved for the IMA of the system? Why or why not?

4

A ramp needs to be installed at a local university's equestrian center to allow wheelchair access to the building. Currently, the height from the ground to the top of the steps into the building is 3 ft. The Americans with Disabilities Act (ADA) requires a 1:12 (or less) ratio for ramp slope. This means that for every inch of height change, at least 12 inches must be run along the base of the ramp.

- Illustrate and annotate the inclined plane described above.
- Determine the minimum allowable length of the ramp base by referring to the given ADA code.
- Using the known height and calculated base length, find the length of the ramp's slope.
- Calculate the IMA of the ramp.
- If a person and their wheelchair have a combined weight of 205 lb, how much ideal effort force is required for them to travel up the ramp?

5

An industrial water shutoff valve in a horse barn is designed to operate with 30 lb of effort force. The valve will undergo 200 lb of resistance force applied to a 1.5 in. diameter axle.

- Illustrate and annotate the wheel and axle system described above.
- What is the system's required actual mechanical advantage (AMA)?
- What wheel diameter is required to overcome the resistance force?

Reflection

Why are you asked to solve for AMA and not IMA?

Building with VEX

6. Form a group with one or two of your peers.
7. Review the **VEX® Parts** and **VEX® V5 Build Support Guides**.
8. Using VEX parts, make three of the simple machines found in the combine harvester.

Turn and Talk

Share your designs with another team in your class. Discuss the differences and similarities in both teams' designs.

HISTORICAL HIGHLIGHT

The Building of the Great Pyramid of Giza

The Giza Pyramids in Egypt were constructed more than 4,500 years ago. The Great Pyramid (Figure 1) is the largest of the three pyramids in Giza and is made of more than 2.3 million blocks of rough, yellow limestone. For the Egyptian people to build the Great Pyramid, they created and used simple machines to help lessen the work.

Workers used wedges to break large pieces of limestone into smaller pieces in the quarry. When the limestone was the correct size, they used levers to move the blocks from the quarry onto wooden sleds. They used inclined planes to move the wooden sleds from the quarry to the building site, which was sometimes more than 500 miles away! The engineering feats accomplished in ancient Egypt are so impressive that their work is studied even today[2].

Figure 1. The Great Pyramid of Giza

Conclusion

Question 1

Create a scenario where a first- or third-class lever would be useful. Include the weights and distances required to solve the problem. When it is complete, give it to a classmate to solve.

Question 2

Who worked on your problem? Were they able to successfully solve it? Were any important pieces of information missing?

Sources

Third-party images, videos, and Historical Highlight information used throughout this activity were drawn from the following sources:

1. How a Combine Works: A view inside the combine [4k video], YouTube, uploaded by Griggs Farms LLC on January 12, 2020, https://www.youtube.com/watch?v=ZMDw9mUoG2M&t=19s, permitted for use by Griggs Farms LLC.
2. Encyclopaedia Britannica, Inc. (2023, January 5). Great Pyramid of Giza. Encyclopaedia Britannica. https://www.britannica.com/place/Great-Pyramid-of-Giza